THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 7: **Work, Energy and Simple Machines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A force of 20 N acts at right angles to the displacement of 5 m. The work done is: 1

(a) 100 J **(b)** 4 J **(c)** zero **(d)** 25 J

2. A lamp of 60 W is used for 5 hours daily for 30 days. The energy consumed in kWh is: 1

(a) 9 **(b)** 300 **(c)** 90 **(d)** 1.8

3. In a lever, the effort arm is 40 cm and the load arm is 10 cm. The MA is: 1

(a) 0.25 **(b)** 30 **(c)** 50 **(d)** 4

4. A skier of mass 60 kg slides down from rest through a vertical drop of 20 m. The speed at the bottom if friction is negligible (g = 10) is: 1

(a) 400 m/s **(b)** 40 m/s **(c)** 20 m/s **(d)** 10 m/s

5. A body of mass 4 kg is lifted through 3 m and then carried horizontally 5 m. The total work done against gravity is (g = 10): 1

(a) 320 J **(b)** 120 J **(c)** 0 **(d)** 200 J

6. The cost of the above energy at 5 rupees per unit is: 1

(a) 45 rupees **(b)** 225 rupees **(c)** 9 rupees **(d)** 450 rupees

7. The effort needed to lift a 200 N load with the above lever is: 1

(a) 20 N **(b)** 200 N **(c)** 50 N **(d)** 800 N

8. A ball dropped from 5 m rebounds to 3.2 m. The fraction of energy lost is: 1

(a) 3.2 per cent **(b)** 64 per cent **(c)** 36 per cent **(d)** 50 per cent

9. The kinetic energy of a body is 50 J and its mass is 4 kg. The velocity is: 1

(a) 12.5 m/s **(b)** 25 m/s **(c)** 3.5 m/s **(d)** 5 m/s

10. A motor lifts 100 kg to 15 m in 30 s. The power is (g = 10): 1

(a) 500 W **(b)** 50 W **(c)** 1500 W **(d)** 5000 W

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A force of 40 N moves a body through 5 m in the direction of the force. Find the work done and the KE gained if the body starts from rest with mass 4 kg. 2

12. A bullet of mass 10 g moving at 300 m/s penetrates 3 cm into a wall. Find the average resistive force. 2

13. Calculate the MA of a lever with load arm 25 cm and effort arm 1 m and the effort required for a 400 N load. 2

14. A man lifts a load of 25 kg to 2 m in 5 s. Find the work and the power (g = 10). 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A 2 kg body is thrown upward with 15 m/s. Find KE and PE at the start, at 5 m and at the top (g = 10). 3

16. A lever has a load of 100 N at 20 cm from the fulcrum. Find the effort at 50 cm and at 1 m on the other side and compare MA. 3

17. Explain why the energy of a falling body is converted to heat when it hits the ground. 3

OR

A crane lifts a 500 kg load through 20 m in 40 s. Find the work done and power (g = 10). 3

18. Explain the use of an inclined plane with a calculation: a 300 N drum is raised by 1 m using a 5 m ramp. Find effort (ideal) and MA. 3

OR

A 20 kg child slides down a slide of height 3 m and reaches the ground at 6 m/s. Find the work done against friction. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A body of mass 2 kg falls freely from 45 m. Find the PE and the KE at the start, after 1 s, 2 s and at the ground and show that the total energy remains constant (g = 10). 5

OR

Explain the efficiency of a machine. A pulley system has VR = 4 and lifts 300 N with 100 N effort. Find MA, efficiency and the energy lost for a 2 m rise of the load. 5

20. Explain the working of a hydroelectric power plant and calculate the power available if 5000 kg of water falls per second through 40 m with 80 per cent efficiency (g = 10). 5

OR

Two pumps lift water to the same height: A lifts 1200 kg in 10 minutes, B lifts 1000 kg in 6 minutes from 20 m depth. Compare their powers (g = 10). 5

END OF QUESTION PAPER — Total: 40 Marks

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